Finite-length and vertical effects in nonlinear-kicker injection into a Korea Light Source design lattice

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Abstract

A single thin kick map can miss trajectory displacement inside a nonlinear injection kicker. For a 0.525 m four-loop magnet in a 4 GeV Korea Light Source design lattice, we compare tracking in an exact filament field, thin and sliced vector maps, and an independent adaptive trajectory solver. A sagitta-to-kick-width ratio organizes finite-length differences over two related wire families; vertical focusing depends on the sampled flank. With the design booster beam and a declared chamber, nominal capture is 97.0 % at 500 turns and 96.1 % at 4000 turns. Under the assumed joint machine/shot budget, mean capture is 14.8 %, conditional on ten corrected rings; these values are not operational efficiency predictions. Ideal centred centroid cancellation also does not ensure full-beam transparency: finite moments change, and wire placement and current imbalance impose separate tolerances. Conditional jitter differences and held-out capture-based tuning quantify the remaining margin without asserting a universal injection tolerance.

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I. INTRODUCTION

Fourth-generation storage rings reduce the electron emittance by an order of magnitude or more with multibend-achromat lattices [1, 2]. The strong focusing and sextupoles that achieve this leave a small dynamic aperture (DA), which becomes the main constraint on injection. These rings also run in top-up mode [3]: the beam is replenished every few minutes during user operation, so each injection should leave the stored beam undisturbed. Where the DA is too small for off-axis injection, the beam can be injected on axis by swapping bunches with an accumulator [4] or injected into the longitudinal acceptance [5]. Where off-axis injection remains possible, the conventional four-kicker orbit bump leaves residual oscillations because the kicker pulses cannot be matched perfectly.

A single pulsed multipole whose field vanishes on the stored-beam axis avoids this. Injection with a pulsed quadrupole was demonstrated at the KEK PF-AR [6] and with a pulsed sextupole at the KEK Photon Factory [7]. A non-linear kicker (NLK) with an off-axis field maximum was developed at BESSY II [8, 9], and a multipole injection kicker (MIK) met a transparency specification of 10 % of the rms beam size at MAX IV and reached 2 % at SOLEIL [10–12]; because commissioning pulsed-multipole injection is not straightforward, a single dipole kicker was added at MAX IV for the commissioning stage [13]. At SIRIUS, NLK injection was predicted to reach 99 % efficiency for a -9 mm DA but delivered about 85 ± 8 % with the design sextupoles, and online sextupole optimisation was needed [14, 15]; BESSY II likewise reached up to 97 % only with optimised sextupoles [9].

The price of transparency is a kick that varies across the injected bunch. Liu *et al.* [14] identified this nonlinearity and the injected-beam size as the main efficiency limit, and placing the bunch near the kick extremum is the natural remedy. Design studies usually describe the kicker by a kick map, the straight-line field integral applied at the magnet centre [16]. A single thin map requires a finite-length accuracy check: the injected beam is deflected by several milliradians inside the magnet, and the vertical kick off the midplane follows from the field's curl-free structure, which requires a consistently signed vector map. Integrating the field along the actual path, as is done for insertion devices and fringe fields [17], resolves trajectory displacement while preserving the physical vector field.

The Korea Light Source (KLS) is a 4 GeV, 28-cell hybrid multibend-achromat ring of about 800 m circumference with a natural emittance of 62 pm [18]. This article studies these

effects for a 0.525 m four-loop wire NKM in a KLS design lattice, with the chamber-limited acceptance reported below. The kicker is modelled by the exact three-dimensional field of its current filaments, validated against the RADIA [19] kick map, and tracked through in slices. The contributions are a quantified finite-length comparison, dimensionless sagitta-to-kick-width scaling over two related wire families, an independent off-midplane trajectory check, and conditional capture and transparency tolerances. The signed vertical field is a consistency requirement for vector maps, not a new deficiency of kick-map methods. The study includes a declared chamber and ten corrected-error rings; it does not predict commissioned-machine performance.

BESSY hardware values describe the 2011 design, whereas the quoted efficiency is from 2023; they are not a single contemporaneous specification. SOLEIL's active compensation and measured construction errors demonstrate why geometric symmetry alone is insufficient [12]. The present tolerance study addresses idealized single-error sensitivities rather than reproducing those machines.

II. SYSTEM AND MODEL

A. Storage ring

The supplied file `K4GSR_HBIv4-1` is identified by the author as the lattice used by Kim *et al.* at IPAC'25 [18]. It is a design version, not the latest KLS lattice; its file identifier is retained because an independent release date was not supplied. Later lattice versions may change the nonlinear acceptance, so results here apply to this version.

The KLS design lattice (file `K4GSR_HBIv4-1`; source circumference 799.30 m, 3481 elements) is tracked with the Accelerator Toolbox [20, 21] in six dimensions with RF and deterministic radiation damping (table II). No stochastic quantum-excitation pass is included. The ring is re-ordered to start at the NKM centre, which is the kick and observation point. The synchrotron period is about 290 turns and the horizontal damping time about 3950 turns. The final capture studies include the declared chamber and septum blade as well as dynamical losses (section V). Figure 1 isolates the supplied lattice and septum without the additional chamber profile: on momentum, particles with $x' = 0$ are stable between -12 and $+13$ mm.

TABLE I. Selected published multipole-injection systems and this simulation. Top: hardware dimensions and a quoted kick (not necessarily a peak). Bottom: acceptance, efficiency and normalized disturbance, with their original conditions. A dash means no comparable value was established from the cited source. Different lattices, observation points and efficiency definitions prevent a performance ranking.

System / type	Magnetic length [m]	Kick [mrad]	Physical aperture [mm]
PF / sextupole [7]	0.300	1.44 at $x = 15$ mm	66 bore diameter
BESSY II / NLK [8]	0.280	1.0 at $x = 12$ mm	42×10
MAX IV / MIK [10]	—	—	—
SOLEIL / MIK [12]	0.304	2.2 nominal	46.8×7.8
SIRIUS / NLK [14, 15]	—	—	—
This study / four-loop NKM	0.525	4.89 peak	60×30 assumed

System	Horizontal DA	Capture / efficiency	Stored-beam disturbance
PF	—	—	—
BESSY II	—	up to 97% with optimized sextupoles [9]	—
MAX IV	—	—	below 10% rms criterion [12]
SOLEIL	$-13/+11$ mm at injection	97% peak, bare lattice	below 2% rms with active compensation [12]
SIRIUS	design -9 mm	99% simulated (213 turns); about 85% measured, 8% variability	—
This study	first lost grid: -11.0/12.5 mm	97.0% simulated (500 turns, reference chamber)	0.10 of σ_x at 0.2 mm horizontal offset, ideal geometry

TABLE II. Principal parameters of the supplied design lattice and study geometry. Booster emittance and energy spread are design values; extraction Twiss, bunch length and vertical ratio are assumptions.

Ring energy, circumference	4 GeV, 799.30 m
Tunes, chromaticities	68.18/23.26, +6.0/ + 3.4
Emittance, energy spread	61.6 pm rad, 1.26×10^{-3}
Optics at NKM centre	$\beta_x = 16.27$ m, $\alpha_x = -0.145$, $\beta_y = 5.60$ m, $D_x = 0.3$ mm
Stored-beam size at NKM	$\sigma_x = 31.7\,\mu$m; $\varepsilon_y = 0.1\,\varepsilon_x$ assumed
Booster extraction Twiss	$\beta_{x,y} = 7.56/12.27$ m
Booster beam at extraction	$\varepsilon_{x,y} = 7.67/0.767$ nm rad, $\sigma_\delta = 1.06 \times 10^{-3}$
RF frequency, total voltage	499.593 MHz, 3.08 MV
Momentum compaction, synchrotron tune	7.77×10^{-5}, 3.44×10^{-3}
Damping times ($x/y/\delta$)	10.54/19.43/16.79 ms
Rms bunch length (stored / injected)	3.63/13.4 mm
Reference arc / centre-bend half-apertures	12×9 / 12×4.87 mm
Assumed NKM / open-IVU half-apertures	30×15 / 12×9 mm
Pulse example	5.3 μs half-sine, injection at peak (analytic)
Transfer line (BTS)	21.79 m, matched optics (peak $\beta_{x,y} = 31.4/50.7$ m)
Septum	stored-beam edge −16 mm, blade 2 mm, 2.26 m upstream of NKM centre
NKM	four loops, wires at $(\pm 5.30, \pm 5.30)$ and $(\pm 10.25, \pm 10.25)$ mm, length 525 mm, 3000 A; peak kick 4.89 mrad at $x = -8.6$ mm ($S = 0.85$)

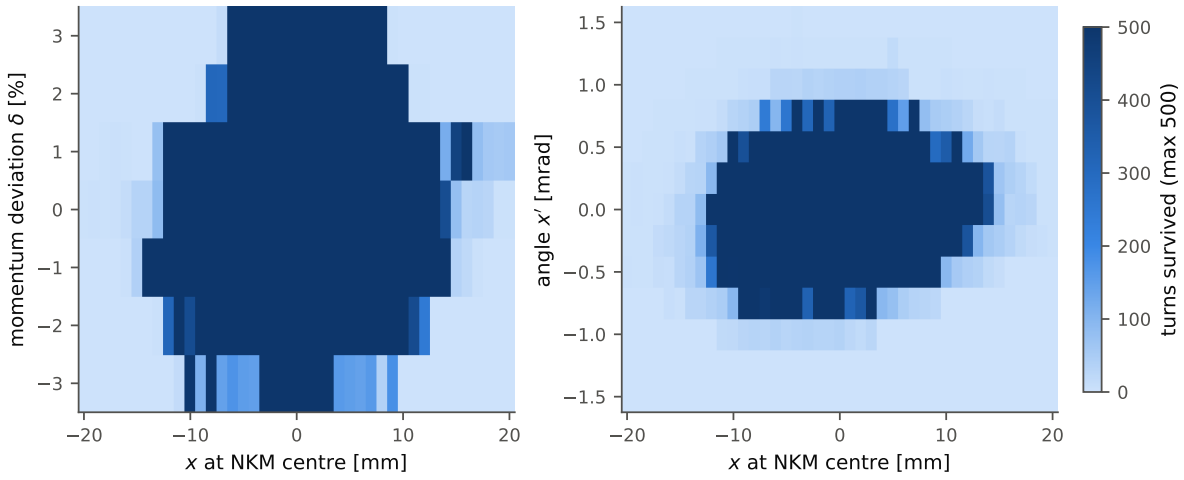


FIG. 1. Deterministic lattice/septum acceptance without the additional chamber profile (no sampling interval), at the NKM centre from native tracking of single particles for 500 turns (1 mm grid in x): turns survived versus x and momentum deviation at $x' = 0$ (left) and versus x and x' at $\delta = 0$ (right).

B. Kicker field and tracking

The 0.525 m four-loop magnet is a preliminary design for this study by Dong-Eon Kim (Pohang Accelerator Laboratory), as stated by the author; it is not the KLS baseline injection kicker. The baseline uses a four-kicker orbit bump and a 1 mm thin septum [22]. The alternative NKM scheme studied here retains an assumed 2 mm blade and a stored-beam edge at −16 mm; these are study geometry, not the published baseline septum. Loop powering has not been specified, so series and independent-loop cases are distinguished in the transparency tolerances.

The NKM consists of four closed rectangular filament loops, one per quadrant, each with an inner and an outer longitudinal wire joined by end connectors (figure 2, left). The field of straight filaments in vacuum is evaluated in closed form (Biot–Savart), including the fringe and end-connector fields. Its straight-line integrals reproduce both components of the RADIA kick map to 7×10^{-12} rad (4×10^{-10} relative) on 3 657 nodes, so the RADIA calculation and the filament model describe the same field. The tabulated map stores $(\int B_y \, \mathrm{d}s, \int B_x \, \mathrm{d}s)/B\rho$; the curl-free kick pair, $\partial \Delta x'/\partial y = \partial \Delta y'/\partial x$, requires $-\int B_x \, \mathrm{d}s$ for the vertical kick. Interpreting the tabulated field component as the vertical kick without the coordinate sign reverses the vertical response; physical thin-map comparisons use the

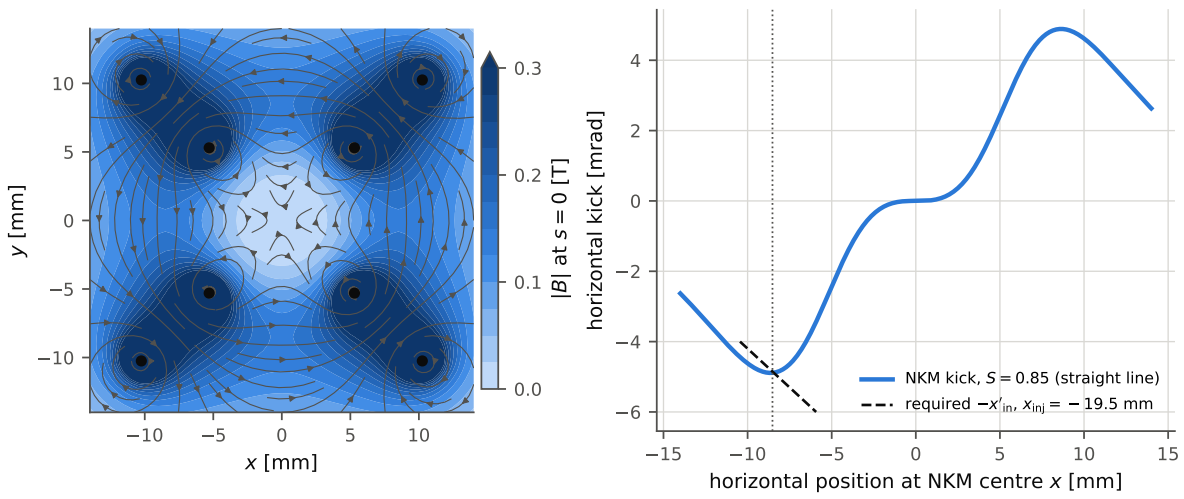


FIG. 2. Deterministic filament-field calculation (no sampling interval). Left: NKM cross-section at $s = 0$ with the wire positions, field lines and $|B|$ of the exact filament field. Right: straight-line horizontal kick at $y = 0$ for $S = 0.85$ and the kick required to cancel the incoming angle of a beam launched from the septum at $x_{\mathrm{inj}} = -19.5\,\mathrm{mm}$; their intersection (dotted) is the thin-kick matched point.

coordinate-consistent sign; the reversed-sign diagnostic is labelled explicitly.

The magnet is tracked in 640 drift–kick–drift slices over $\pm 0.45\,\mathrm{m}$ in canonical AT coordinates, $\mathrm{d}p_x/\mathrm{d}s = PS(B_y - y'B_s)/B\rho$ and $\mathrm{d}p_y/\mathrm{d}s = PS(-B_x + x'B_s)/B\rho$ with slopes $x' = p_x/(1+\delta)$, polarity P and field scale S. The kick acts on canonical momentum and carries no $1/(1+\delta)$ factor; the exit angle changes by $0.12\,\mu\mathrm{rad}$ per 10^{-3} in δ. Slice and span refinement change the exit angle by less than $7 \times 10^{-11}\,\mathrm{rad}$. Particles within 1 mm of a conductor are counted as lost. The polarity is chosen so that the kick at $x < 0$ cancels the angle of a beam travelling from the septum towards the axis (figure 2, right).

An independent adaptive DOP853 integration checks the stated paraxial equations at the nominal point, the final angular-window edges (offsets -0.517/0.241 mrad), nonzero vertical position and angle, and $\delta = \pm 0.002$. Relative/absolute solver tolerances are $2 \times 10^{-11}/2 \times 10^{-13}$. Maximum 640-slice discrepancies are $3.26 \times 10^{-11}\,\mathrm{m}$ in x, $5.08 \times 10^{-10}\,\mathrm{rad}$ in p_x, $2.90 \times 10^{-11}\,\mathrm{m}$ in y and $5.06 \times 10^{-10}\,\mathrm{rad}$ in p_y. Slice and span checks through 1280 slices and $\pm 0.9\,\mathrm{m}$ are below one percent of a 0.1σ response in either plane. The field is exact for the filament geometry; tracking remains a validated paraxial approximation.

TABLE III. Independent trajectory check. Launch y is in mm, y' in mrad and δ in 10^{-3}; absolute coordinate discrepancies are nm for positions and nrad for canonical momenta. These are deterministic solver comparisons, without sampling intervals.

Case	y	y'	δ	$\lvert\Delta x\rvert$	$\lvert\Delta p_x\rvert$	$\lvert\Delta y\rvert$	$\lvert\Delta p_y\rvert$
nominal	0.0	0.0	0.0	0.024	0.137	0.000	0.000
lower angle	0.0	0.0	0.0	0.011	0.508	0.000	0.000
upper angle	0.0	0.0	0.0	0.033	0.139	0.000	0.000
vertical position	1.0	0.0	0.0	0.025	0.226	0.027	0.500
vertical angle	0.0	0.2	0.0	0.024	0.137	0.002	0.004
both vertical	1.0	0.2	0.0	0.024	0.226	0.029	0.505
$\delta < 0$	1.0	0.2	-2.0	0.024	0.227	0.029	0.506
$\delta > 0$	1.0	0.2	2.0	0.024	0.225	0.029	0.504

C. Injected beam, capture and controls

The injected bunch is generated at booster extraction and tracked element by element through the matched BTS line to the septum exit, where it is placed at $(x_{\mathrm{inj}}, x'_{\mathrm{inj}})$. It propagates to the magnet, is tracked through the 3-D NKM once (single-turn pulse, section VIII) and then for 500 turns. *Capture* is the fraction of injected particles alive after the last turn; intervals are 95 % Wilson intervals [23] for particles and cluster-bootstrap intervals for error realizations. Lost particles are classified by upstream transport, septum creation, incoming transport, magnet domain, return septum or elsewhere in the ring; the categories reconcile with the particle count. The *matched* angle is the inner root of zero centroid angle after the magnet. Two controls receive identical particles: a uniform *dipole* kick and a *linearised* kick (with its curl-free vertical partner), both calibrated to the 3-D response of the nominal centroid. This angular calibration does not reproduce every finite-length position–angle correlation, so it cannot uniquely decompose the capture difference into a nonlinear-kick contribution.

III. FINITE-LENGTH AND VERTICAL EFFECTS

At a diagnostic launch $x_{\mathrm{inj}} = -20\,\mathrm{mm}$ and $S = 0.85$, the thin kick map gives a matched angle of 4.870 mrad, the 3-D magnet 4.6475 mrad. To isolate the cause, the map's horizontal profile was distributed over the magnet length and sampled either along the undeflected straight line or along the self-consistent deflected path (figure 3). Straight-path sampling gives 4.762 mrad; the deflected path gives 4.6476 mrad, equal to the 3-D result, and reproduces its response to angle errors to 0.01 mm. Straight-path sampling already shifts the thin-map match, and self-deflection produces an additional shift. Its sagitta is $|K|L/8 = 0.32\,\mathrm{mm}$. Agreement between the deflected-profile and 3-D calculations supports this mechanism in the tested configuration; it does not make longitudinal sampling or end fields universally irrelevant.

The effect is asymmetric. A positive septum angle error moves the trajectory inwards; on the straight line it would cross the kick extremum onto the weaker inner flank, leaving the extra angle under-compensated. The self-deflection instead holds the trajectory on the outer side of the extremum, where the kick is largest (figure 3, centre). A +0.3 mrad error changes the betatron amplitude of the kicked centroid by +1.22 mm with the thin kick but by -0.12 mm with the 3-D magnet, whereas −0.3 mrad gives +0.66 and +0.60 mm; the nominal amplitude is larger (9.57 against 9.07 mm). At fixed integrated kick, a longer magnet lowers the sensitivity to positive errors but raises the nominal amplitude, and for $L \geq 0.75\,\mathrm{m}$ no matched solution remains (figure 3, right). Small sagitta relative to the kick-lobe width supports a thin-map approximation; a sliced vector map can resolve the deflection when that ratio is appreciable.

At this diagnostic launch the 3-D magnet defocuses vertically: a centroid at $y = 1\,\mathrm{mm}$ receives 0.25 mrad, against 0.13 mrad from the sign-corrected thin map, because the deflected path samples the field off the straight line. For paired 100 nm rad horizontal-emittance sensitivity bunches at this diagnostic launch, capture is 91.1 % with a reversed vertical-kick sign, 89.7 % with the corrected sign and 81.8 % with the 3-D magnet. Suppressing only the 3-D vertical kick restores 87.8 % (the horizontal-only finite-length model gives 87.4 %), so at that point the vertical defocusing costs about five percentage points. For a curl-free thin vector kick, the linear vertical coefficient is $-\partial\Delta x'/\partial x$, with flank-dependent focusing sign, and vanishes at the kick extremum; at the operating point selected in section IV, where the

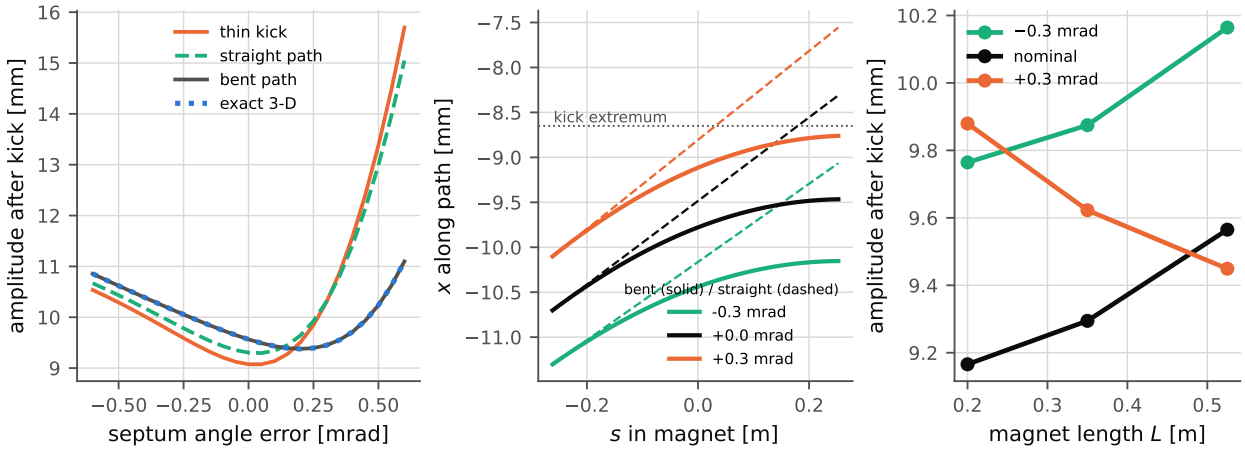


FIG. 3. Deterministic finite-length comparison at the −20 mm diagnostic launch; no sampling interval. Left: betatron amplitude of the kicked centroid at the NKM centre versus septum angle error for the thin kick, the profile sampled on the straight and on the deflected path, and the exact 3-D field, each at its own matched angle. Centre: centroid path inside the magnet for angle errors of −0.3, 0 and +0.3 mrad (solid: deflected; dashed: straight) relative to the kick extremum. Right: deflected-path model versus magnet length at fixed integrated kick.

centroid sits near the extremum, the approximately two-point cost is a 100 nm rad sensitivity result (section V), not the design-beam headline.

The sagitta ratio $\eta = |K|L/(8w)$ compares trajectory displacement with the horizontal kick-lobe full width at half maximum w. Figure 4 varies length, integrated strength and a second wire family scaled transversely by 1.25. It is a deterministic model comparison, not a universal capture criterion. A sliced vector map retains self-deflection and the physical vertical kick; the local focusing sign changes between flanks of the kick extremum. A reversed vertical-kick sign is an input-convention error, not a failure inherent to kick maps. Capture controls throughout are paired model comparisons on identical particles, not independently optimized injection systems.

IV. OPERATING WINDOW

At the final point and reference chamber, the native ring captures 97.0 % at 500 turns, versus 100.0 % for an affine element-map reference on identical particles. The reference retains element lengths and explicit aperture checks at physical boundaries, and linearizes about the

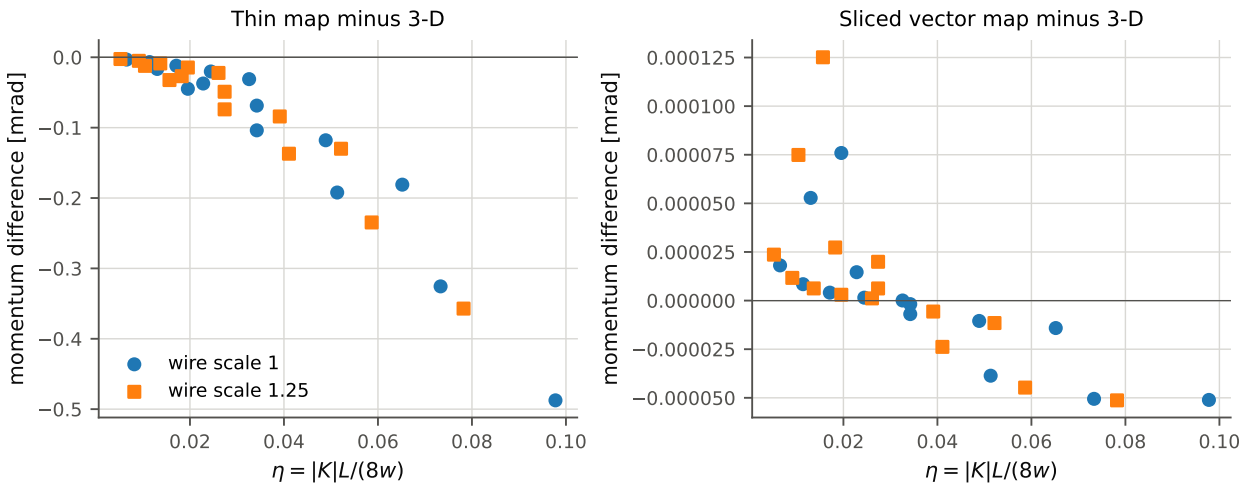


FIG. 4. Single-centroid momentum differences versus η for five lengths (0.2–1.0 m), three integrated strengths and two transverse wire families. Left: thin vector map minus 3-D field. Right: vector map distributed over 1280 slices minus the 3-D field. Width w is the negative kick lobe's full width at half maximum. The centroids have nonzero vertical position and angle; these deterministic comparisons have no sampling interval.

six-dimensional closed orbit with RF and radiation. A small-perturbation one-turn check agrees within 1.36×10^{-14} in the corresponding metre/radian/dimensionless coordinates. It deliberately omits nonlinear dynamics; all capture conclusions use native tracking.

Nominal native survival at 4000 turns is 96.1 % (95 % particle Wilson interval 94.7–97.1 %), a loss of 0.9 percentage points after turn 500. The 500-turn prefix reproduces the paired chamber-study particle masks exactly. Fresh bunches at the interpolated lower and upper angular-edge candidates give 79.5/85.0 % at 500 turns and 79.4/84.7 % at 4000 turns (95 % intervals 76.8–81.8 % and 82.3–86.8 %). The degraded corrected ring 5501 gives 3.0 % at 500 turns and 1.4 % at 4000 turns (0.8–2.3 %). Thus the stated window remains a 500-turn conditional contour, not an infinite-time acceptance or a guaranteed 4000-turn 80 % contour.

The point $x_{\mathrm{inj}} = -19.5\,\mathrm{mm}$, $S = 0.85$, $x'_{\mathrm{inj}} = 4.853\,\mathrm{mrad}$ was selected by the largest particle-Wilson lower bound in a fixed matched-point grid using a 100/10 nm rad sensitivity beam. Four fresh bunches gave 93.0 % (91.7–94.0 %) for that beam and septum-only aperture. This independent sample reduces reuse of selection noise; it does not remove dependence on the searched grid or model. The design-beam/chamber studies retain these

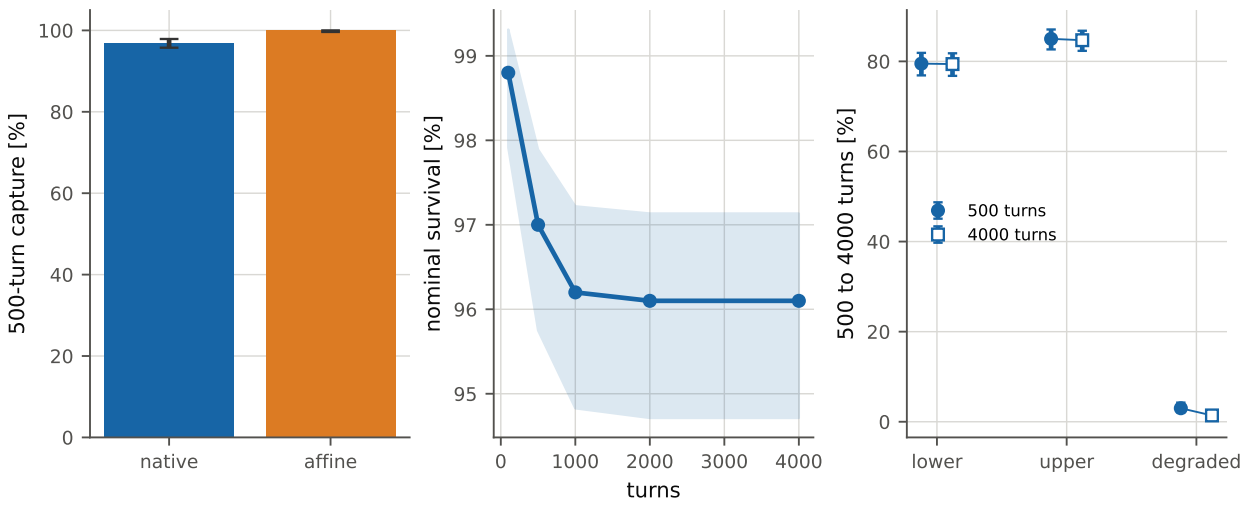


FIG. 5. Final-point validation with the 3-D NKM, design beam and reference chamber. Left: native and affine-element backends at 500 turns on the same two 500-particle bunches (seeds 1001/1002; 95 % particle Wilson bars). Centre: native nominal survival with turn count (same particles; Wilson band). Right: 500-to-4000-turn survival for interpolated lower/upper angular-edge candidates and corrected ring 5501, each with two fresh 500-particle bunches (seeds 2001/2002); circles/squares denote 500/4000 turns, with 95 % Wilson bars. Intervals are conditional on the selected models and ring, not machine-population uncertainty.

knobs without reselection and therefore test a conditional operating point rather than a new global optimum.

We estimate the connected operating window above 80 % capture around that fixed point. Linear interpolation gives angle offsets from -0.52 to 0.24 mrad (width 0.76 mrad), and position offsets span -0.67 to 1.32 mm (width 1.99 mm). Each value uses seeds 1001/1002 and 500 particles per seed. A fixed 0.2 mm/mrad grid is extended if a boundary remains above threshold, with a maximum offset of 3; crossing cells are refined once to 0.1 spacing and linearly interpolated. Grid resolution is not a confidence interval for a crossing. The long-turn edge checks use fresh particles and do not redefine the selected 500-turn window.

V. SURVIVAL, ACCEPTANCE AND SENSITIVITY

A larger-emittance sensitivity case uses 100/10 nm rad and rms energy spread 0.0011, with the septum aperture but without the additional chamber profile. For 1000 particles

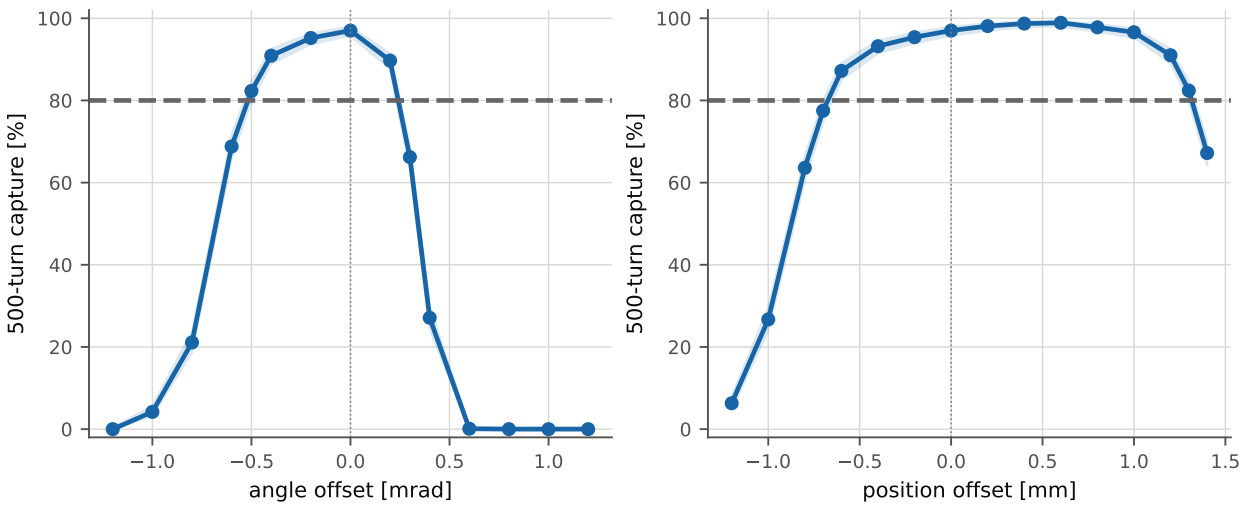


FIG. 6. Design-beam operating window with the reference chamber and fixed NKM settings. Left/right: capture versus septum-exit angle/position offset, 500 native turns, two 500-particle bunches (seeds 1001/1002) per point. Bands are conditional 95 % particle Wilson intervals; the dashed line marks 80 %. Crossing cells have 0.1 mrad/mm spacing after refinement.

(seeds 1001, 1002), 93.0 % survive 500 turns and 89.5 % survive 4000 turns (95 % particle Wilson interval 87.4–91.3 %). The late loss is 3.5 percentage points (figure 7); this number belongs to the larger-emittance case and is not the design-beam long-turn result. A separate 2000-particle sample gives 92.4 % (91.2–93.5 %) at 500 turns.

In that sensitivity case, the paired vertical-kick diagnostic gives 93.0 % with the NKM vertical kick and 94.8 % when it is suppressed. A uniform 5 mm vertical half-aperture reduces capture with or without that kick (to 86.3 and 88.0 %). This supports a role for transverse transfer in the ring, without diagnosing a unique coupling resonance. These artificial uniform-aperture scans are separate from the final chamber model.

A separate selected corrected-ring scenario applies correlated 30 μm girder offsets, 30 μm independent magnet offsets, 100 μrad rolls, gradient/sextupole errors and skew-quadrupole components, all explicit sensitivity assumptions. Ten seeds (5501–5510) use the 120 HVC correctors and BPM markers with a truncated-SVD response, followed by bounded Q11/Q12 tune correction. The selected rms criterion is 100 μm; a stricter 20 μm target was not attained. Maximum achieved BPM rms residuals are 84.6/56.2 μm, and the largest fractional-tune residual is 9.1×10^{-6}. The full DA grids (figure 8) define the subsequent chamber/error scenario; they are not a commissioned-machine prediction.

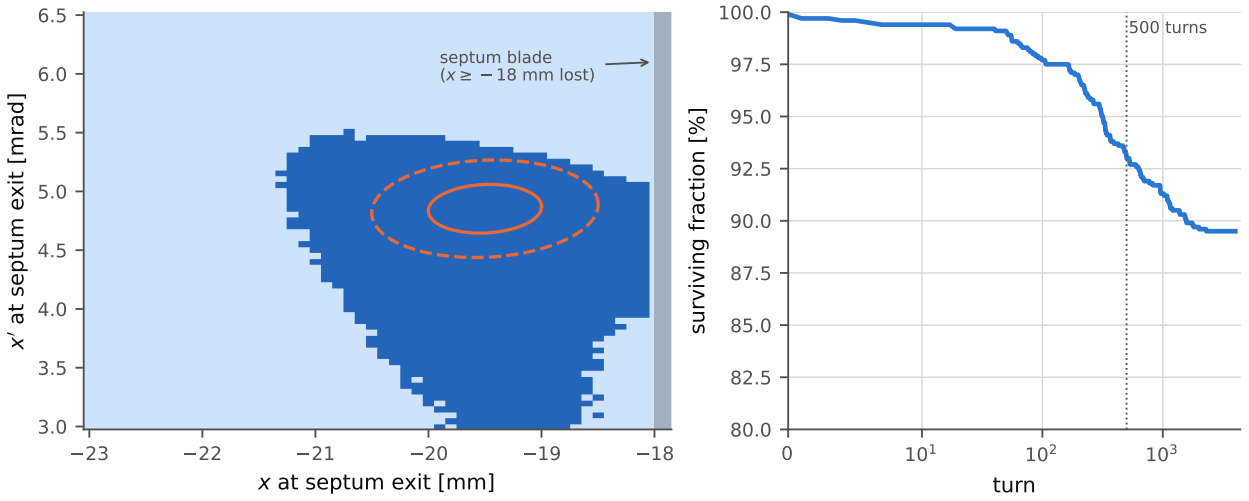


FIG. 7. Larger-emittance sensitivity case without the additional chamber profile. Left: deterministic 500-turn single-particle survival in septum phase space ($\delta = 0$, 3692 launches), with 100/10 nm rad beam ellipses at one and two rms widths. Right: survival of 1000 particles, seeds 1001, 1002, through 4000 native turns. The line is the pooled particle fraction; its final 95 % Wilson interval is stated in the text. The deterministic acceptance grid has no sampling interval.

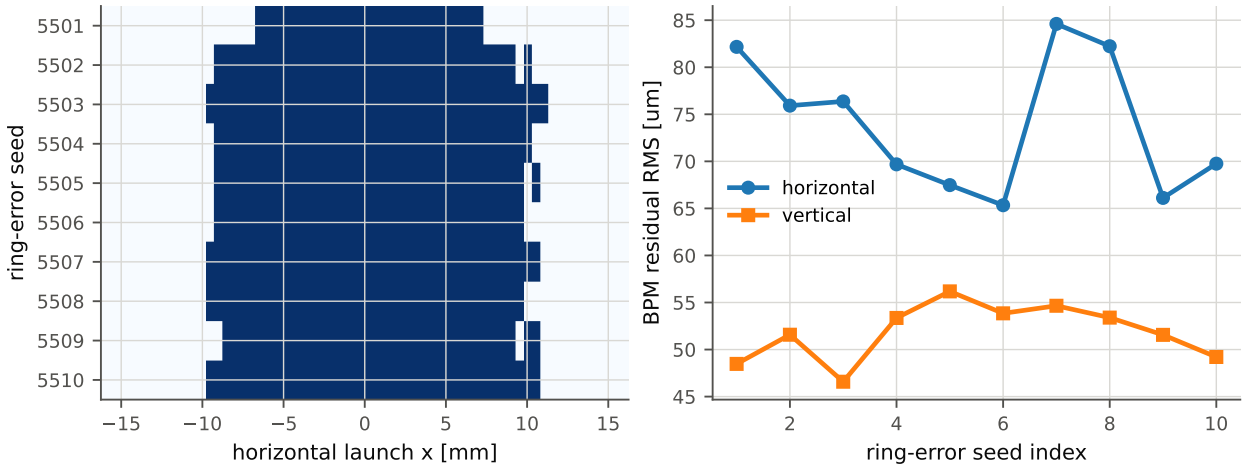


FIG. 8. Selected corrected-ring scenario, ten fixed seeds. Left: deterministic 500-turn native survival on a 0.5 mm horizontal launch grid, with RF/radiation and the reference chamber. Blue/white cells denote surviving/lost launches. First failed grid points are not exact stable edges. Right: horizontal and vertical BPM rms residuals after SVD orbit and bounded tune correction. Curves show seed variation, without a particle-sampling interval.

The reference arc half-apertures are 12 mm horizontally and 9 mm vertically [24, 25]. The centre-bend vertical half-aperture is 4.87 mm. An inscribed ellipse represents the unspecified octagonal corners. An assumed rectangular NKM straight has half-width/height 30/15 mm. Nineteen identified ordinary straights receive 3 m IVU segments [25]; their assignment is a scenario, not a surveyed map. The reference open half-gap is 9 mm; a 2.5 mm closed magnetic half-gap is a stress case and does not account for a liner. The separate incoming transport retains its source lattice and septum creation cut; its chamber has not been surveyed for this alternative injection scheme.

For the 7.67 nm rad design booster beam, the reference chamber gives 97.0 % capture at 500 turns (95 % particle Wilson interval 95.7–97.9 %; two 500-particle bunches, seeds 1001/1002). The paired dipole gives 97.3 %; the NKM-minus-dipole difference is -0.3 percentage points (95 % paired-particle bootstrap interval -0.9 to +0.3; 10 000 draws, seed 1729). In rectangular arc scans, half-widths 10, 12 and 15 mm give 81.7, 97.0 and 99.7 % at each tested half-height (4.87, 7.5, 9 mm). Thus 10 mm is the smallest tested rectangular-grid width exceeding 80 % in this ideal-ring sample, while 12 mm retains the reference result; neither is a continuous minimum or a machine tolerance. Across the ten corrected rings, mean capture is 61.7 %, median 63.7 %, tenth percentile 52.3 %, and range 2.7–87.4 %. These are different descriptors of a finite scenario population.

VI. STORED-BEAM TRANSPARENCY

The kick of the four-loop magnet is odd in x on the midplane and its near-axis expansion has a leading cubic term, $\Delta x' \simeq c_3 x^3$ with $c_3 = 2.4 \times 10^4\,\mathrm{rad\,m^{-3}}$ at $S = 0.85$. For an aligned kicker and a centred, symmetric Gaussian beam the ideal centroid kick therefore vanishes exactly. This does not imply unchanged finite-beam moments: deterministic quadrature gives single-pass projected emittance changes of 4.5×10^{-7} horizontally and 1.6×10^{-7} vertically. Orders 20, 40 and 60 per axis agree at the reported precision. The meaningful quantities are the response to an offset between kicker and stored orbit and the emittance change, evaluated deterministically from the exact field by Gauss–Hermite quadrature (figure 10, right). Offsets of 0.1, 0.2 and 0.5 mm give centroid amplitudes A_x of 0.016, 0.10 and 1.54 of the rms size, and the spread of the kick across the beam increases the emittance about the displaced centroid by 9.4×10^{-5}, 1.2×10^{-3} and 4.2×10^{-2}; a 0.2 mm vertical offset

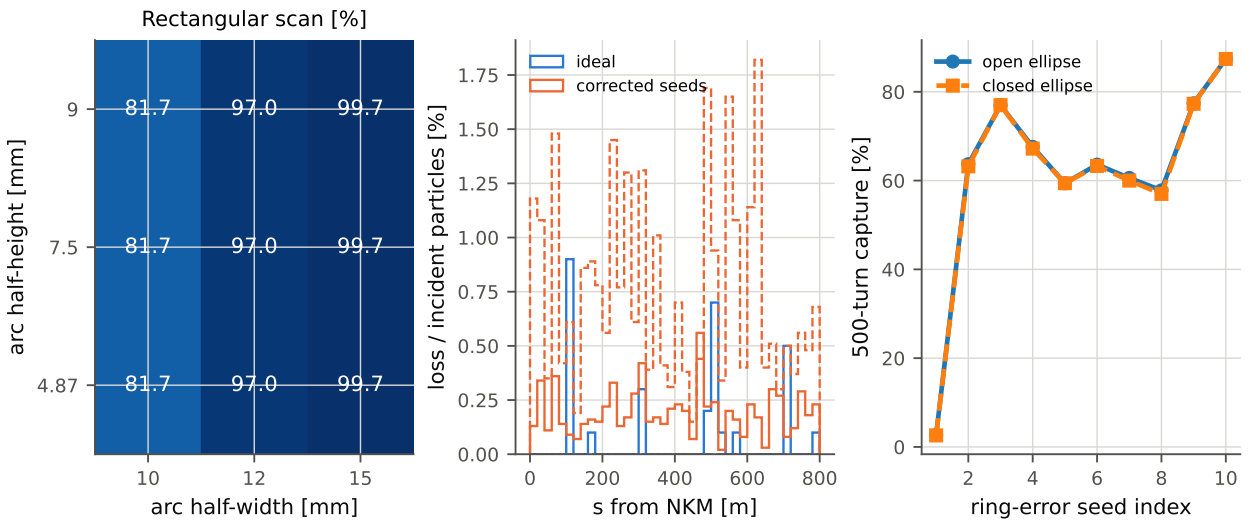


FIG. 9. Declared chamber with the design booster beam, 500 native turns and two seeds of 500 particles per ring/profile/model. Left: ideal-ring capture versus rectangular arc half-apertures, with other region limits retained. Centre: ring losses per incident particle in 20 m bins using element-entrance positions from the NKM centre; solid/dashed/dotted denote horizontal/vertical/dynamical losses. For elliptic apertures the larger normalized coordinate defines the plane. Right: capture for each corrected ring with open and closed IVUs. Points are conditional sample fractions; the loss histograms and finite ring-seed variation have no assigned confidence band.

gives 0.17 of the vertical rms size. Once the centroid oscillation has also filamented, the rms emittance grows by the factor $1 + A_x^2/2 + \Delta\varepsilon_x/\varepsilon_x$, i.e. 1.007 at 0.2 mm and 2.22 at 0.5 mm. Native tracking uses two seeds (11 and 12), each with 300 paired particles, with radiation on. The maximum centroid difference is measured at the ends of turns 1–50, and projected emittance at the end of turn 1000: $A_x/\sigma_x = 0.10$ and 1.47, and mean paired emittance ratios 1.01 and 2.24. These tracked observables need not equal a fully filamented analytic estimate. In this ideal geometry and at the assumed emittance, an offset of 0.2 mm keeps the stored-beam disturbance at about a tenth of its size, whereas 0.5 mm is not transparent. The bicubic interpolation of the map agrees with the exact field to 9×10^{-11} rad near the axis, whereas bilinear interpolation errs by 1×10^{-6} rad and overstates the 0.1 mm response nineteen-fold (0.29). The calibrated dipole control would displace the stored beam by about 2495 rms sizes in a linear estimate; this hypothetical figure only illustrates why a uniform kicker cannot be used for top-up.

The assumed vertical-to-horizontal emittance ratio is 0.10, as supplied by the author; for

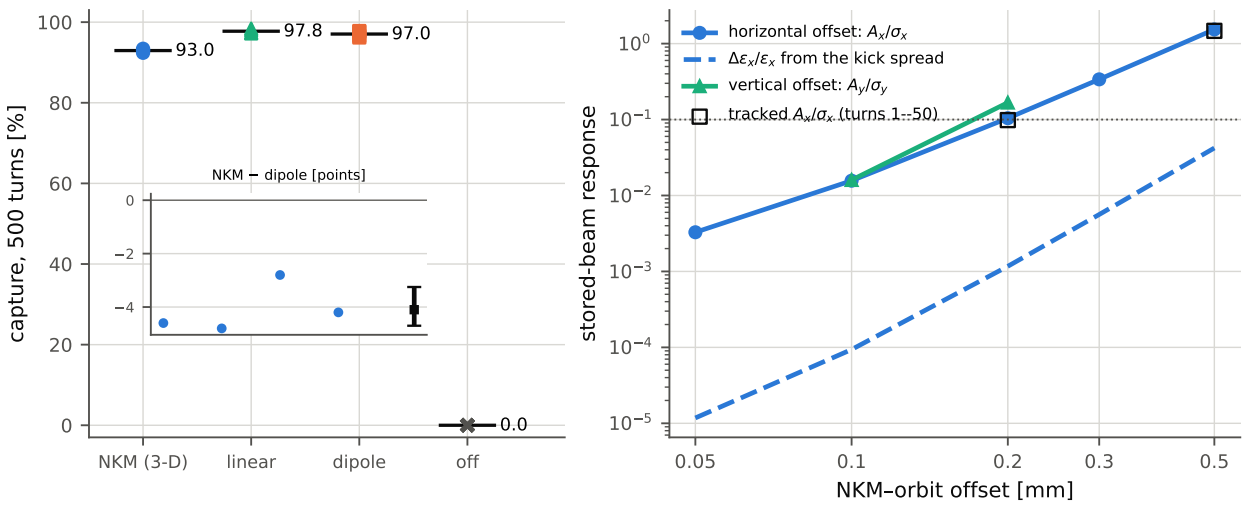


FIG. 10. Left: 100/10 nm rad sensitivity case with the septum aperture and no additional chamber; four fresh 500-particle bunches at the selected point for the 3-D NKM, the calibrated linearised and dipole controls and the kicker off (markers: bunches; bars: mean). The inset shows per-bunch NKM-minus-dipole differences (dots), their mean (square) and a paired-bunch bootstrap 95 % interval (10 000 resamples, seed 1729; only four bunches). Right: deterministic centroid amplitude and filamented emittance growth versus NKM-to-orbit offset. Open squares are the mean of two 300-particle tracked seeds, using maximum centroid differences over ends of turns 1–50; they are not 1000-turn amplitudes. Emittance ratios in the text use the end of turn 1000. No sampling interval is assigned to deterministic quadrature.

a fixed vertical dipole kick, $A_y/\sigma_y \propto \varepsilon_y^{-1/2}$. Individual wire displacements and independent loop-current errors break the cancellation (tables IV and V). The one-error-at-a-time wire limits range from 0.29 to 2.04 μm for both centroid amplitudes to remain below 0.1 σ. Independent loop currents require relative balance of about 1.1×10^{-4}. In series powering, a common current error preserves dipole cancellation provided the geometry remains symmetric; geometry errors still matter. These are local linear tolerances, not an allocation for simultaneous errors or a manufacturing specification.

The paired controls (figure 10, left) compare the selected models: the linearised and dipole kicks capture 97.8 and 97.0 %, the paired NKM-minus-dipole difference is -4.1 points (95 % paired-bunch bootstrap interval -4.7 to -3.2), conditional on this larger-emittance sample; with the kicker off no particle survives.

The published baseline thin-septum leakage of about 3.3 G at the bumped orbit [22],

TABLE IV. On-axis wire sensitivity. Wire numbers follow the four quadrant loops, inner then outer wire. u is the displaced coordinate. Dipole derivatives $D_{x,y} = \partial K_{x,y}/\partial u$ are in rad/m; $Q_n = \partial J_{xx}/\partial u$ and $Q_s = \partial J_{xy}/\partial u$ are in rad/m^2, with $J = \partial(K_x, K_y)/\partial(x, y)$, $J_{yy} = -J_{xx}$ and $J_{yx} = J_{xy}$ to numerical precision. t_x, t_y are single-error 0.1-sigma limits in μm. A dash indicates no resolved first-order dipole sensitivity; higher orders remain. Deterministic central differences, step 0.1 μm; no statistical interval.

Wire	u	D_x	D_y	Q_n	Q_s	t_x	t_y
1	x	-0.000	+0.357	-67.4	+67.4	–	0.29
1	y	+0.357	-0.000	+67.4	+67.4	0.54	–
2	x	+0.000	-0.096	+9.3	-9.3	–	1.10
2	y	-0.096	+0.000	-9.3	-9.3	2.04	–
3	x	-0.000	-0.357	+67.4	+67.4	–	0.29
3	y	-0.357	-0.000	+67.4	-67.4	0.54	–
4	x	+0.000	+0.096	-9.3	-9.3	–	1.10
4	y	+0.096	+0.000	-9.3	+9.3	2.04	–
5	x	-0.000	+0.357	+67.4	-67.4	–	0.29
5	y	+0.357	-0.000	-67.4	-67.4	0.54	–
6	x	+0.000	-0.096	-9.3	+9.3	–	1.10
6	y	-0.096	+0.000	+9.3	+9.3	2.04	–
7	x	-0.000	-0.357	-67.4	-67.4	–	0.29
7	y	-0.357	-0.000	-67.4	+67.4	0.54	–
8	x	+0.000	+0.096	+9.3	+9.3	–	1.10
8	y	+0.096	+0.000	+9.3	-9.3	2.04	–

treated as uniform over 0.4 m, gives $\Delta x' \simeq \int B_y\, ds/(B\rho) = 9.9\,\mu$rad at 4 GeV ($B\rho = 13.34$ T m). If applied locally at the NKM optics, this would be an order-five-σ_x response. This is a scale estimate, not the field at the unbumped NKM stored orbit, which is farther from the blade. The shielded thick-septum value 0.3 ± 0.1 G over 1.7 m similarly corresponds to approximately $3.8 \pm 1.3\,\mu$rad per magnet before accounting for geometry and transport. A local leakage map, shielding and correction assessment are needed before claiming trans-

TABLE V. Independent-loop relative-current sensitivity. Dipole derivatives are in rad per relative current; normal and skew quadrupole derivatives are in rad/m per relative current. The final column is the joint two-plane 0.1-sigma single-loop limit in units of 10^{-4}. Deterministic derivatives with relative-current step 10^{-5}; no statistical interval.

Loop	D_x	D_y	Q_n	Q_s	Limit
1	-0.0009	-0.0009	+0.000	-0.262	1.15
2	+0.0009	-0.0009	+0.000	+0.262	1.15
3	+0.0009	+0.0009	+0.000	-0.262	1.15
4	-0.0009	+0.0009	+0.000	+0.262	1.15

parency of the entire injection system; the NKM-only cancellation does not supply that evidence.

VII. ROBUSTNESS AND TOLERANCES

A. Error budget

Table VI defines the final untruncated Gaussian rms budget. Booster horizontal emittance 7.67 nm rad and energy spread 0.106 % are design values [26]; extraction Twiss, bunch length and the 0.10 vertical-emittance ratio are assumptions. No public extraction-orbit or septum-current stability was supplied, so the corresponding errors remain sensitivity parameters. Small exit Twiss mismatch is not used as evidence of capture robustness.

The published bump-kicker reproducibility is not treated as an NKM or septum specification. At the lattice RF frequency, phase error shifts the injected longitudinal coordinate by $c\Delta\phi/(2\pi f_{\mathrm{RF}})$. Independent arrival relative to NKM firing scales the 5.3 μs half-sine field at peak phase and does not add another longitudinal displacement. The separate thin-septum error adds canonical momentum $(1+\delta)\,0.01\,\epsilon_{\mathrm{sept}}$ in radians. No equivalent transverse timing-kick term is added. Longitudinal quadrupole placement, supply quantization and a scalar ring-beta perturbation are not modelled and are not counted as applied errors.

TABLE VI. Final rms error budget. S: static machine draw; J: shot draw. Unsourced entries are explicit sensitivity assumptions with the tested effect stated. Corrected-ring errors are a separate finite scenario population.

Term	RMS	Class	Source or reason
Booster centroid x/x'	0.5 mm / 0.2 mrad	J	Extraction stability unavailable; scan scales 0–1
Booster beta mismatch	5%	S	Assumed extraction-optics sensitivity
Emittance / mean energy	10% / 0.1%	J	Assumed fluctuations about design beam
Energy-spread variation	10^{-4} absolute	J	Assumed variation of within-bunch spread
BTS quadrupole gradient	0.1%	S	Assumed focusing error
BTS quadrupole x/y offset	100 μm each	S	Assumed alignment error
BTS quadrupole roll	0.5 mrad	S	Assumed coupling error
BTS dipole field	0.05%	S	Assumed rigidity/trajectory sensitivity
Septum position	0.1 mm	S	Assumed blade geometry error
NKM / residual orbit offset	0.2 mm each	S	Assumed relative alignment sensitivity
Common NKM amplitude	0.5%	J	Assumed pulser fluctuation, not a measured specification
Thin-septum strength	0.1% relative	J	Assumed; nominal bend 10 mrad [22]
RF phase	0.001 rad	J	Assumed injection-phase sensitivity
Arrival relative to NKM firing	2 ns	J	Assumed peak-phase pulse timing

B. Combined errors and tolerance curves

Three seeds (42, 123, 777) generate 150 joint machine/shot realizations, each with 200 Gaussian particles and 500 native turns. Each realization independently draws one of the ten corrected rings with replacement, plus the errors in table VI. The bootstrap unit is the joint realization, conditional on that finite ring population; it is not an estimate over all possible commissioning errors or repeated shots on one machine. Intervals use 10 000 percentile-bootstrap resamples, seed 1729.

Ideal static steering transports the zero-input reference through the perturbed BTS, giving (r_x, r_{p_x}), then translates the bunch at the septum by $(x, p_x) \mapsto (x - r_x, p_x - r_{p_x})$. The other coordinates are unchanged. This removes no shot jitter, vertical trajectory error or optics distortion, and does not model a fitted orbit-correction system.

The 3-D NKM mean is 14.8 % (95 % interval 11.2–18.8 %), median 1.0 %, tenth percentile 0.0 %, and fraction of realizations at or above 80 % is 2.7 %. The paired dipole mean is 22.5 % (18.3–27.0 %). The NKM-minus-dipole mean difference is -7.7 percentage points (-10.3 to -5.4). These controls compare models on identical particles and errors; they are not separately optimized injection systems.

The jitter scan retains the same 50 seed-42 realizations, particle seeds and ring identities while scaling only booster position and angle jitter. Table VII reports differences from zero jitter. Quarter jitter corresponds to 0.125 mm and 0.05 mrad; its paired mean difference is +2.2 points, with interval -1.5 to +6.1. The interval does not establish the predeclared one-percentage-point loss margin; the candidate jitter target remains precision-limited. Position-only and angle-only reference jitter are evaluated separately. Mean, median, tenth percentile and fraction above threshold remain distinct; no 80 % design specification follows from a nominal ideal-ring value.

Partial Spearman tests control the other sampled errors and ring-identity indicators. Their asymptotic two-sided t-test probabilities receive Holm adjustment across the sampled-error hypotheses. Scalar errors enter as magnitudes and vector errors as rms values. The smallest Holm-adjusted probabilities below 0.05 correspond to booster angle ($\rho = -0.51$, $p_{\mathrm{Holm}} = 2.20 \times 10^{-8}$). These describe conditional associations, not causal error rankings; the complete coefficient and adjusted-probability table is included in the numerical evidence.

For reduced capture-based tuning, each of the 50 seed-42 static machines receives 100 fresh

TABLE VII. Paired jitter sensitivity, 50 joint realizations per row. Capture summaries and the fraction of realizations at or above 80 % are percentages; differences from zero jitter and their paired 95 % bootstrap intervals are percentage points.

Scale	Mean	Median	p10	Fraction ≥ 80%	Difference 95% interval
0	34.1	28.2	0.0	4	+0.0 [+0.0, +0.0]
0.25	36.3	31.5	0.0	6	+2.2 [-1.5, +6.1]
0.5	29.4	24.0	0.0	4	-4.7 [-9.6, +0.0]
0.75	22.9	7.0	0.0	4	-11.2 [-17.8, -5.2]
1	19.8	3.5	0.0	4	-14.3 [-21.3, -7.8]
position only	30.8	21.5	0.0	6	-3.3 [-8.5, +1.5]
angle only	20.8	3.8	0.0	6	-13.2 [-19.6, -7.2]

training particles with shot fluctuations disabled. A single bounded coordinate sweep evaluates seven settings of $(x_{\mathrm{inj}}, x'_{\mathrm{inj}}, S)$: bounds $[-22, -18.2]$ mm, $[3.5, 6.5]$ mrad and $[0.7, 1.2]$, with steps 0.5 mm, 0.3 mrad and 0.05. Strict improvements are accepted and ties retain the current setting. The fixed seven-evaluation budget is the termination rule; settings remain feasible within those bounds. Independent held-out 200-particle samples then restore the joint shot errors at zero and quarter booster jitter. The tuned-minus-untuned gains are +11.6 points (+7.8 to +15.5) and +5.5 points (+1.1 to +10.0), respectively, with paired 95 % realization-bootstrap intervals. The corresponding tuned means are 45.7 and 41.8 %; fractions of realizations above 80 % are 12.0 and 10.0 %.

Training capture has one-percentage-point counting resolution. No BPM calibration, resolution, charge-monitor noise or actuator-error model is included, so this reduced ideal-observable search quantifies recovery without demonstrating operational beam-based feedback. Held-out results are never used to choose knobs, and the dipole control is not retuned.

VIII. PULSED-MAGNET CONSIDERATIONS

The results above apply to an ideal single-pass spatial field. Native tracking of the nominal centroid observes positions -2.9, 6.2 and 9.0 mm after turns 1–3 at the NKM centre. The corresponding straight-line full-field kicks are 0.54, 3.71 and 4.87 mrad. Assigning one tenth

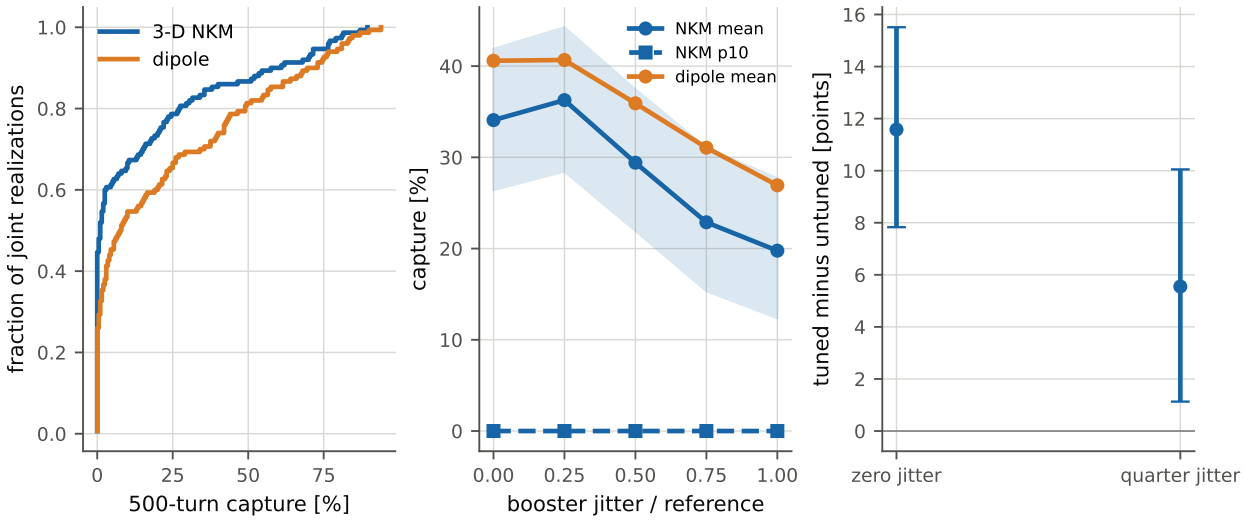


FIG. 11. Final design-beam robustness with the reference chamber. Left: empirical cumulative capture distributions for 150 paired joint-error draws, conditional on ten corrected rings. Centre: NKM mean and tenth percentile and dipole mean versus jitter scale (50 paired draws); shading is a 95 % realization-bootstrap interval of the NKM mean. Right: held-out tuned-minus-untuned mean capture at zero and quarter jitter, with paired 95 % intervals (50 machines, 100 training and 200 held-out particles each). Bootstrap count 10 000, seed 1729. All tracking uses 500 native turns.

of the conditional angular-window width (0.076 mrad) as a residual allowance is a design assumption; it gives first/second-return fractional limits 0.14/0.02.

For a half-sine of total duration τ, injection at its peak ($t = \tau/2$) gives first-return field fraction $\cos(\pi T_0/\tau)$ if $T_0 \le \tau/2$, and zero otherwise. Later returns replace T_0 by nT_0 in this finite-support rule. The native-centroid allowance gives $\tau \le 5.85\,\mu$s; zero field on every return requires $\tau \le 5.33\,\mu$s. A $5.3\,\mu$s peak-phase example meets the latter condition, whereas $8\,\mu$s retains 0.50 of the peak on the first return. Deterministic timing offsets obey $1 - \cos(\pi\Delta t/\tau)$, giving 7.7×10^{-8}–4.9×10^{-6} for 1–5 ns and 5–8 μs. The 2 ns Gaussian rms in the error model is a distribution, not a hard timing bound.

A train with its first bunch at the peak samples less field on later bunches: for the $5.3\,\mu$s example, 100, 200 and 500 ns train durations give approximately 0.18, 0.70 and 4.4 % variation. At an offset kicker the stored-beam centroid response scales with that field; aligned centroid cancellation does not imply zero finite-beam moment change. On-axis residual dipole and gradient allowances are 4×10^{-5} and 7×10^{-5} of the corresponding peak quantities. A hypothetical thin cylindrical conducting shell has diffusion time $\tau_{\mathrm{eddy}} \simeq \mu_0\sigma a t/2$: at

$a = 15\,\mathrm{mm}$, a $1\,\mu\mathrm{m}$ coating with $\sigma = 10^6\,\mathrm{S/m}$ gives 9.4 ns, whereas a 1 mm wall with $\sigma = 5 \times 10^7\,\mathrm{S/m}$ gives 0.47 ms. These are order-of-magnitude scenarios, not predictions for the unspecified NKM chamber. Coating-induced multipoles require chamber-specific analysis [11]; no time-dependent capture or operational pulse feasibility is claimed.

IX. DISCUSSION AND LIMITATIONS

Finite-length accuracy is controlled by trajectory displacement relative to the kick-lobe width, not by the label "kick map". The tested sagitta ratio organizes deterministic comparisons over a scaled wire family; it does not establish a universal capture threshold. A sliced vector map retains both the physical vertical field and self-deflection. Local vertical focusing changes sign across the kick extremum. The observed transfer of large horizontal oscillations into vertical losses is consistent with coupled nonlinear ring dynamics, without identifying a unique resonance.

Nominal capture, a joint-error mean and the fraction of realizations above 80 % answer different questions. The mean and lower tail depend strongly on the declared corrected-ring population. The 500-turn window is a conditional survival contour; its 4000-turn checks quantify later losses. Ideal BTS steering removes two static coordinates only, while the reduced tuning search uses an ideal capture observable. Neither establishes a practical feedback tolerance or commissioned efficiency. Small BTS Twiss mismatch alone does not establish successful ring capture.

The ring version, aperture shape, IVU assignments, incoming-line clearance, error budget and ten corrected rings limit transfer to hardware. The filament model omits iron, chamber-induced fields, pulse evolution and actual septum leakage. The pulse section supplies consistency estimates, not time-dependent capture. Sextupole optimization may alter the acceptance, as commissioning studies at BESSY II and SIRIUS illustrate [9, 15]; it is outside this study. The joint capture ensembles vary whole-magnet placement and common field scale, not individual wires or loop-current imbalance. Wire tolerances refer to one error at a time, and aligned centroid symmetry does not eliminate finite-beam moment changes or manufacturing errors.

X. CONCLUSIONS

For the supplied KLS design lattice, the 3-D filament NKM and design booster beam give 97.0 % nominal capture with the reference chamber at 500 turns and 96.1 % at 4000 turns, a late loss of 0.9 percentage points. The interpolated conditional 80 % angular-window width is 0.76 mrad. Finite-length displacement and consistently signed vertical fields require validation, while a sliced vector map provides a controlled comparison within the tested configurations. The final joint-error mean is 14.8 %, conditional on ten corrected rings; reduced tuning and paired jitter statistics quantify its sensitivity rather than define a universal 80 % specification. Ideal aligned centroid cancellation coexists with finite-beam moment changes and micrometre-scale wire tolerances. Pulse and septum estimates further limit any claim of system-level transparency.

DATA, SOFTWARE AND CODE AVAILABILITY

The simulation code, configurations and analysis scripts are available in the nkm-injection repository (`https://github.com/cspark7701/nkm-injection`). The figures and numbers are generated from the campaign artifacts listed, with SHA-256 digests, in `figure_provenance.csv` of the manuscript directory `docs/paper/generated`. A frozen evidence bundle with the selected simulation outputs, input files, code snapshot and rebuild instructions is prepared as a referee attachment with a hash inventory and verified rebuild instructions. It has no public DOI; the repository alone does not contain the ignored simulation outputs.

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the content.

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